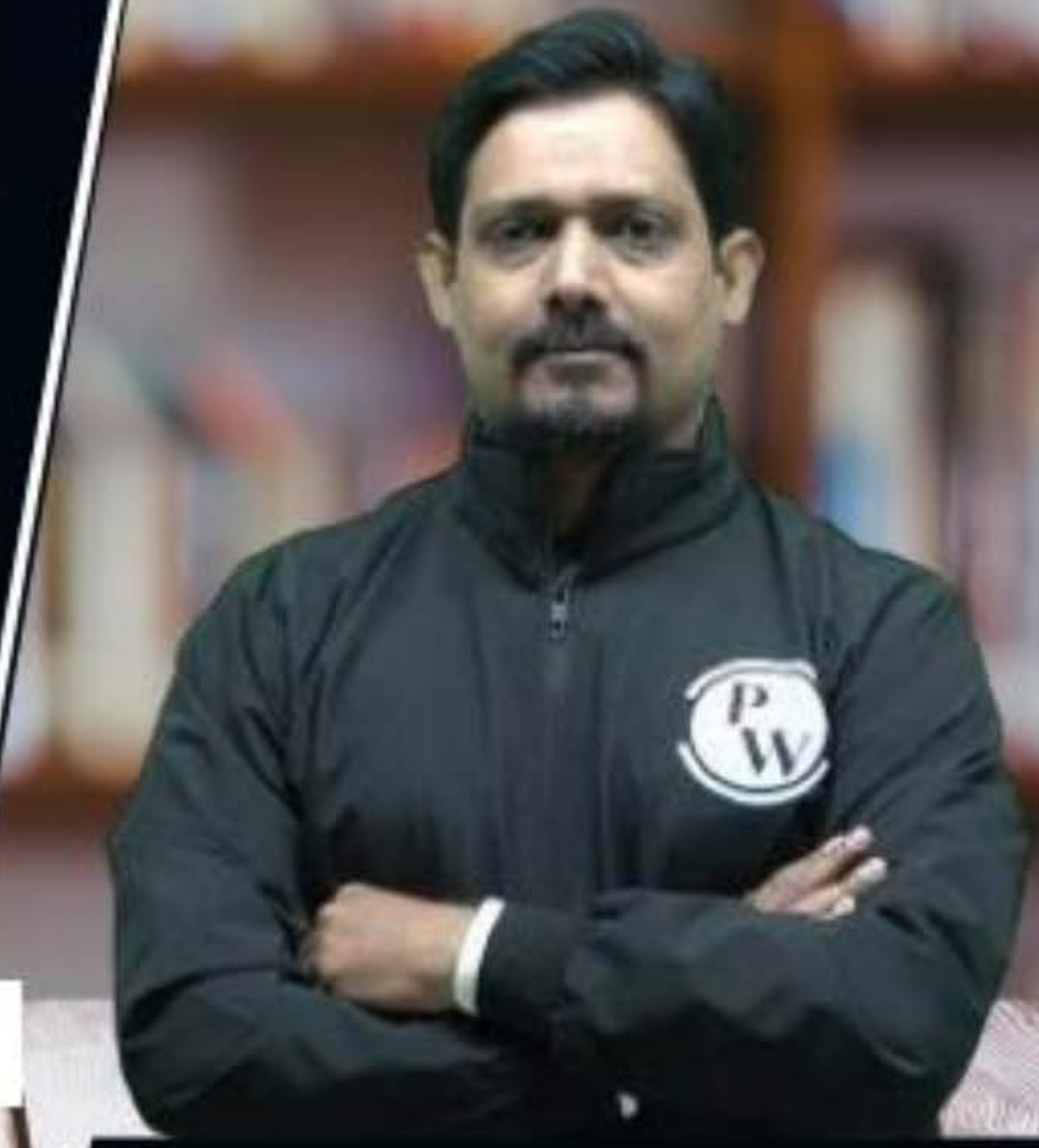
Computer Science & IT

Data Structure & Programming

Stack

Lecture No. 02



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Recap of Previous Lecture

Topic — Stack as LIFO

Topic — Stack implementation

Topic — Stack as permutation generator.

Topic

Topic

Slide

Topics to be Covered

Topic — Expression in Computer

Topic — Infix, prefix, postfix

Topic — Conversion using precedence & using stack

Topic

Topic

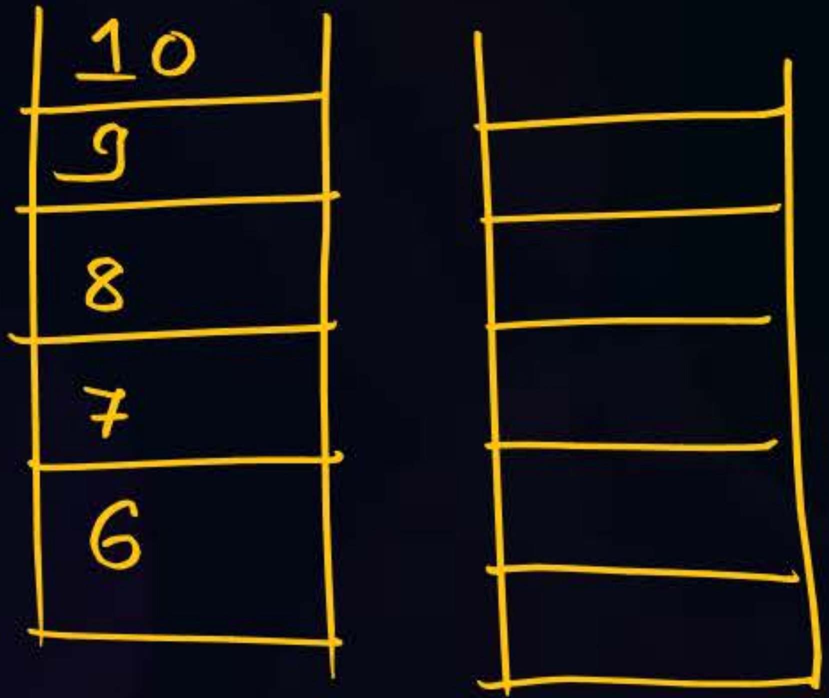
Slide

Question

10

Suppose the Push operation takes 2 units of time, Pop takes 1 unit, and Printing an element takes 3 units of time. Two stacks A1 and A2 are used:A1 initially contains the elements {6, 7, 8, 9, 10} (with 10 on the top),A2 is initially empty. What is the minimum time required to print the sequence:10, 8, 6, 9, 7

Push :- 2 units
pop :- 1 unit.
printing :- 3 units



pop (A_1) print
pop (A_1) push(A_2)

Slide

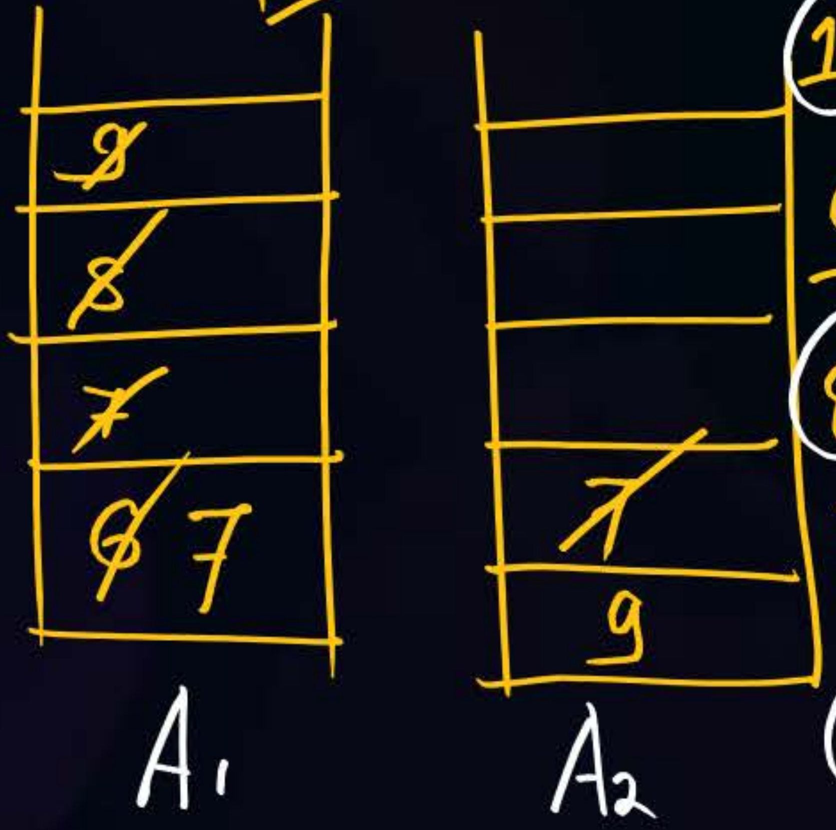
Question

No restriction (

10, 8

Suppose the Push operation takes 2 units of time, Pop takes 1 unit, and Printing an element takes 3 units of time. Two stacks A1 and A2 are used:A1 initially contains the elements {6, 7, 8, 9, 10} (with 10 on the top),A2 is initially empty. What is the minimum time required to print the sequence:10, 8, 6, 9, 7

pop



10 pop (A_1) print

9 pop(A_1) push(A_2)

8 pop(A_1) print

7 pop(A_1) push

6 pop(A_1) print

7 pop(A_2) push(A_1)

pop(A_2) print 9

pop(A_1) print 7

Push :- 2 units × 3 = 6

pop :- 1 unit × 8 = 8

printing :- 3 units × 5 = 15

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Slide

Topic : Expression In Computer

Infix : $a+b$

operand : variable/Numeric value

operator

prefix : $+ab$ polish Notation

postfix : $ab+$ Reverse Polish Notation

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Topic : Expression In Computer

why prefix and postfix

Infix $a * b \hat{} c$: To evaluate we need to identify which operator to evaluate using precedence and Associativity

prefix and post fix expression already precedence and associativity Rule in incorporated

Single scan will evaluate the express ion

Slide

Topic : Expression In Computer

$\underline{-5}$ ← unary minus

a - b → Binary minus

Bracket	( )		Highest
Unary minus	-		
Exponentation	↑	Right to Left	
Multiplicative	* / %	Left to Right	
Additive	+ -	Left to Right	
Equal	=	Right to Left	Lowest

Slide

Topic : Expression In Computer

$-b$ prefix $\rightarrow -b$
postfix $\rightarrow b-$

$A - B/C * D \uparrow E$

4 2 3 1

prefix :

operand

$A - B/C * \uparrow DE$

$A - /BC * \uparrow DE$

$A - */BC\uparrow DE$

$\boxed{-A*/BC\uparrow DE}$

postfix :

$A - B/C * DE\uparrow$

$A - BC/ * DE\uparrow$

$A - BC/DE\uparrow *$

$\boxed{ABC/DE\uparrow * -}$

Slide

Topic : Expression In Computer

$A \underset{3}{+} B \underset{2}{*} C \underset{4}{-} D \underset{5}{-} E \underset{1}{\uparrow} F \underset{6}{+} G$

Convert to prefix

$A + \underline{*BC} - D - \underline{\uparrow EF} + G$

$\boxed{+ - - + A * BC D \uparrow EF G}$

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Topic : Expression In Computer

2004 PYQ.

Assume that the operators +, –, ×, are left associative and ^ is right associative. The order of precedence (from highest to lowest) is ^, x, +, –. The postfix expression corresponding to the infix expression a + b × c – d ^ e ^ f is

(A) abc × + def ^ ^ –

(B) abc × + de ^f ^ –

(C) ab + c × d – e ^ f ^

(D) – + a × bc ^ ^ def

a+b×c-d^e^f

a+ bc* – d^ ef^

a+bc* - def^^

$\boxed{abc*+def\text{^^}-}$

Slide

$-b$ prefix $= -b$

postfix $= b-$

The **prefix** expression for the infix expression

$a = \underline{-b} + \underline{c * d}/e + f \uparrow \underline{g \uparrow h} - \underline{i * j}$

(a) $= a - + + - b / * cde \uparrow f \uparrow gh * ij$

(b) $= a - + + - / * bcde \uparrow f \uparrow gh * ij$

(c) $= a + - + - b / * cde \uparrow f \uparrow gh * ij$

(d) $= a - + - - b / * cde \uparrow f \uparrow gh * ij$

$a = -b + \underline{*cd}/e + f \uparrow \underline{\uparrow gh} - *ij$

$a = -b + \underline{/*cde} + \underline{\uparrow f \uparrow gh} - *ij$

$a = - + + -b / * cde \uparrow f \uparrow gh * ij$

$\boxed{= a - + + -b / * cde \uparrow f \uparrow gh * ij}$

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Topic : Expression In Computer

#Q The infix expression A op1 (B op2 C) op3 D is correctly represented in prefix notation as op1 A op3 op2 B C D, where op1, op2 and op3 are operator . What is the possible value of op1 op2 and op3.

A. op1 = '*', op2 = '*', op3 = '+'

B. op1 = '+', op2 = '*', op3 = '/' ✓

C. op1 = '^', op2 = '*', op3 = '*'

D. None of these

$A\ op1\ (B\ op_2\ C)\ op_3\ D$

$A\ op_1\ \underline{op_2 B C}\ \underline{op_3}\ D$ precedence of $op_3 > op_1$

$A + \underline{op_2 B C} / D$

$A + \underline{op_3\ op_2 B C D}$

$op_1\ A\ op_3\ op_2 B C D$

Topic : Expression In Computer

#Q. Choose the equivalent prefix form of the following expression

(a + (b − c))* ((d − e)/(f + g − h)) = a+ -bc

(a) * +a − bc /− de − +fgh ✓ [A] = (+a -bc) * (-de / -+fgh)

(b) * +a −bc − /de − +fgh

(c) * +a − bc /− ed + −fgh = (+a-bc) * /−de−+fgh

(d) * +ab − c /− ed + −fgh X

= *+a-bc/−de−+fgh

Question

$6 - 2 * 4 + 5$

$6 - * 2 4 + 5$

What is the prefix expression of the given postfix expression 6 2 4 * 5 + -

(A) - 6 + * 2 4 5 ✓ [A]

(B) - + 6 * 2 4 5

(C) + - 6 * 2 4 5

(D) - + 5 * 2 4 6

$\boxed{ab+} \Longrightarrow a+b$

$\boxed{ab+} \Rightarrow +ab$

6 2 4 * . 6 2*4 , 5 +

6 $\boxed{*24, 5 +}$

6 , $\boxed{+ * 2 4 5}$ -

- 6 + * 2 4 5

6, 2, 4 *

put bracket In Infix

6, (2*4) , 5 +

6, ((2*4) +5) -

6 - ((2*4)+5)

Slide

Question

What is the postfix expression of the given prefix expression?

$\leftarrow$ RtoL

+ 8 - + 7 * 2 3 * 4 5

(A) 8 7 2 3 * + 4 5 * - + ✓ (A)

(B) 8 7 2 * + 3 4 * - 5 +

(C) 8 7 2 * + 3 4 5 * - +

(D) 8 7 * 2 + 3 4 * - 5 +

* 4 5

* 2, 3, 45 *

\+ 7, 2 3 * . 45 *

\- 7, 2 3 * +, 45 *

\+ 8. 7. 2, 3 * + . 45 * -

8, 7, 2, 3 * + 45 * - +

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Topic : Question

PYQ : 2015

The attribute of three arithmetic operators in some programming language are given below.

OPERATOR	PRECEDENCE	ASSOCIATIVITY	ARITY
+	High	Left	Binary
−	Medium	Right ✓	Binary
*	Low	Left	Binary

The value of the expression $2 - 5 + 1 - 7 * 3$ in this language is 9.

$2 - 5 + 1 - 7 * 3$

$2 - \underline{6 - 7} * 3$

$2 - (-1) * 3$

$3 * 3 = \underline{9}$

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Topic : Expression In Computer

PYQ 2024 IG

Consider the operator precedence and associativity rules for the *integer* arithmetic operators given in the table below.

Operator	Precedence	Associativity
+	Highest	Left ✓
-	High	Right
*	Medium	Right
/	Low	Right

The value of the expression $3 + 1 + 5 * 2/7 + 2 - 4 - 7 - 6/2$ as per the above rules is ____ [6]

$\boxed{3+1+5} * 2/\boxed{7+2} - 4 - 7 - 6/2$

$9 * 2/9 - 4 - \underline{7 - 6}/2$

$9 * 2/9 - 4 - 1/2$

$9 * 2/9 - 3/2$

$9 * 2/6/2$

$18/6/2$

$18/3 = \underline{6}$

Slide

Infix to Postfix: $a + b * c$

a+bc*
abc*+

Symbol	Stack					Output
a						a
+	+					a
b	+					ab
*	+	HPU				
	+	*				ab
c	+	*				abc
	END OF INPUT pop everything & print					abc*+

Slide

Infix to Postfix : a * b − c

ab* − c

$\boxed{ab*c-}$

Symbol	Stack	Output
a		a
*	*	a
b	*	ab
−	* LPOPU-S	
	−	ab*
c	−	ab*c
	END OF INPUT pop everything & print	ab*c−

Slide

Infix to Postfix: $a * b \uparrow c - d$

Symbol	Stack					Output
a a						a
*	*					a
b	*					ab
↑	*	HPU				
	*	↑				ab
c	*	↑				abc
−	*	↑	LPOPU			abc
	*					abc↑
						abc↑*
	−					abc↑*
d						abc↑*d
	END OF INPUT					abc↑*d−
Slide						

Question

Tomorrow again Discuss.

Last question

If we use a stack to convert the following infix expression to postfix:

a + b * c - d / f * a

What will be the top TWO contents of the operator stack (first element is on the top) immediately after scanning the second * (the one after f)?

(A) * , /

(B) * , -

(C) / , -

(D) - , +

a		a
+	+	
b	+	ab
*	+ *	ab
c	+ *	abc
-	+ *	abc *
-	+	abc *
-	-	abc * +
d	-	abc * + d
/	- /	ab * + d
f	- /	ab * + df
*	- /	ab * + df
	- *	ab * + df /

Incomplete.

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Infix to Postfix using stack

1. Scan input character left to right.

2. Output character left to right .

3. If the input is an operand then print it.

4. If input is the operator and stack is empty then push the ~~operand~~ operator.

5. If input is the * operator having higher precedence than operator + on top of stack the perform push Operation (HPU).

HPU · Higher precedence perform push

Slide

Infix to Postfix using stack

6. If the input operator is having $\boxed{\text{lower precedence}}$ than operator on the top of stack then pop $\boxed{\text{operators}}$ until input ~~operand~~ operator is having lower precedence than operator on top of stack and push new operator. LPOPU-S

Lower precedence
pop operators
push New operator

7. Left Associative LPOPU-S

8. Right Associative HPU

Slide

THANK - YOU

Evaluation of
poefix

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